Home Reading 08 — Nyquist Stability and Margins

Before Class

Preview Lecture 8

Read Lecture 8: Nyquist Stability and Margins.

Textbook Reading

- Required: Ogata, System Dynamics, 4th ed., Section 11-4, pp. 630–632 and 636–637 (Nyquist plots and stability margins).
- Page numbers above are printed textbook pages. Use the Lecture slides for the remaining worked examples.

This Week Introduces

Nyquist stability counting, gain and phase margins, delay budgets, uncertainty and the limits of nominal evidence.

Tutorial Preparation

Read Tutorial 8 before class. Be ready to:

- Q1: Combine a complete Nyquist trace with the stated open-loop pole count.
- Q2: Calculate gain and phase margins and the phase loss from the specified delay. The delay-budget extension is optional.

Check Your Understanding Before Class

Before class, try to answer these two questions:

- Why is the number of unstable open-loop poles needed for a Nyquist stability decision?
- Why does a time delay reduce phase margin even when the plant gain is unchanged?

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